Finding the Fastest Explosions with *TESS*

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Fast transient events have durations ranging from just a few seconds to days. *TESS* has a wide FoV, and high cadence, providing a good opportunity for serendipitous observations of rapid optical transients. Determining the efficiency of our detection pipelines allows us to estimate rates of recoverable transients in *TESS* data.

MOTIVATION

The TESSELATE pipeline conducts an untargeted transient search in *TESS* data, and currently has three main methods to detect candidate events. To test our methods, we can inject ePSF-like sources (shown on the right) into *TESS* data at random positions and times, and then run our detection pipelines.

- ★ **Correlation Method** finds candidates by determining how similar their shape is to the ePSF.
- ★ **Aperture Method** places circular apertures on the centres of candidates, determines the flux, and discards those that have a large FWHM or excessive flux.
- ★ **CNN Method** comes from the SourceDetect pipeline, which uses machine learning to find ePSF-like sources.



Gaussian PSF

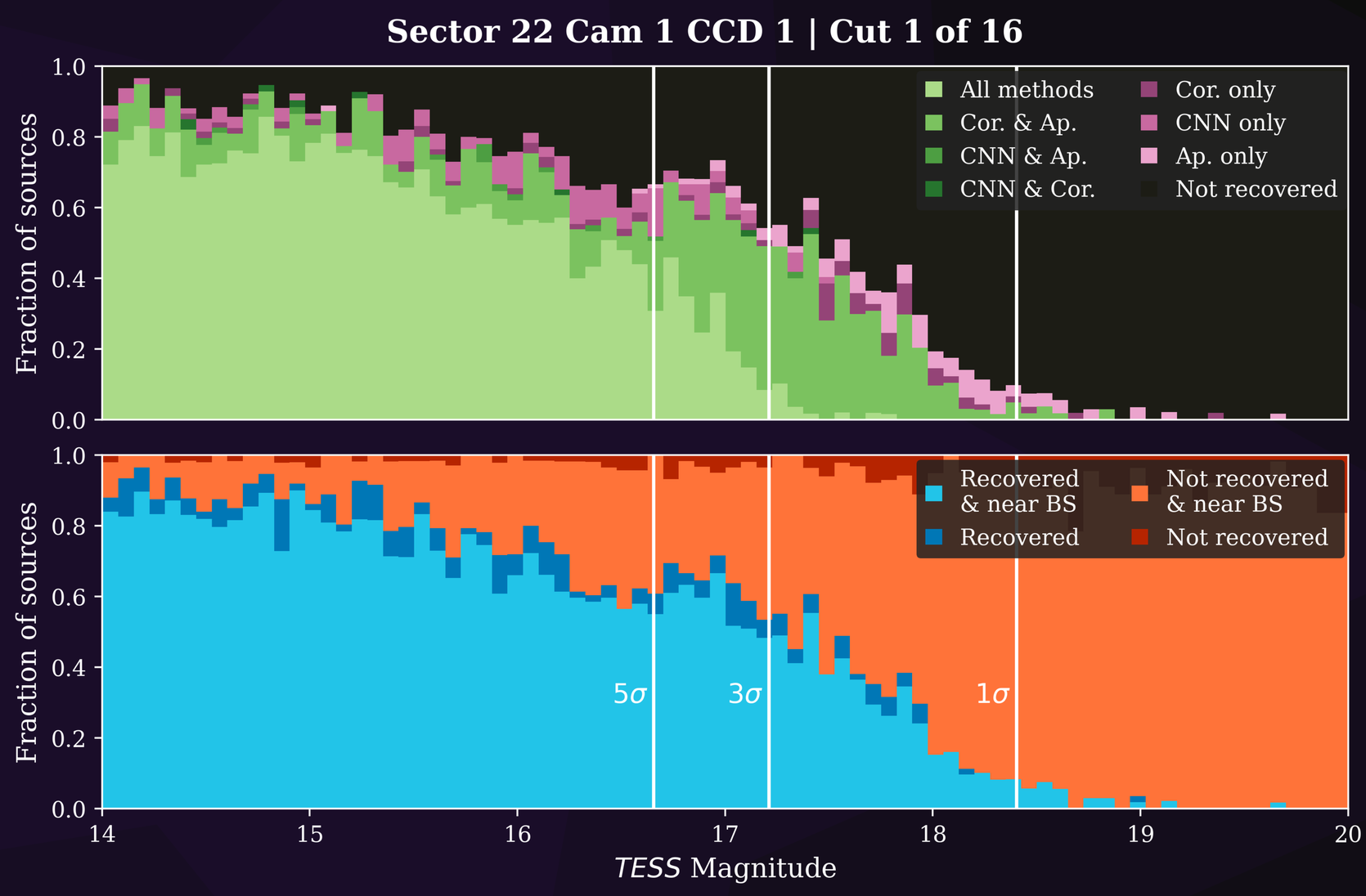
TESS ePSF

DETECTION LIMITS

The efficiency trials showed that *TESS* consistently recovers around 80% of all sources brighter than the typical 5σ limit, with the 1σ limit tending to be $m_{TESS} \sim 18.5$.

The recovery rates varies significantly, both spatially and temporally, most often due to the variability of the background.

Testing the efficiency means we also need to account for bright sources (BS) that may be near our candidate. Within the efficiency trials, if a BS is within a 1-pixel radius of our injected source, this results in the source being flagged.



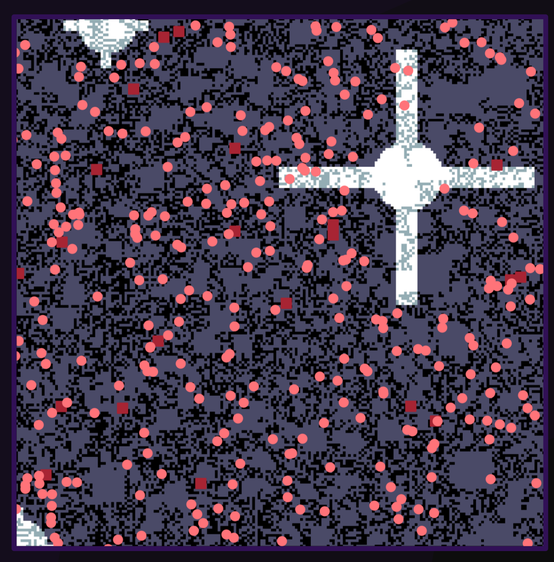
HOW MUCH CAN *TESS* SEE?

The efficiency of *TESS* can be used to determine the recoverable quantity of specific transients we would expect *TESS* to observe. Using the solid angle, Ω, of a cut, we can determine the volume of transients that should appear within the *TESS* FoV using the equation

$$V = \frac{\Omega}{3}\left(10^{\frac{1}{5}(m-M+5)}\right)^3$$

where m is the magnitude limit of the cut and M is the absolute magnitude of the transient.

Assuming a rate for kilonovae of $1540^{+3200}_{-1220}\ Gpc^{-3}yr^{-1}$ with an absolute magnitude of $m_{TESS} \approx -16$ and the FELT rate as $6400 \pm 2400\ Gpc^{-3}yr^{-1}$, with an absolute magnitude that ranges from $\mathrm{m_{TESS}} \approx -16.5$ to -20 [1].



A cutout of the recovered ePSFs shown on top of the *TESSreduce* mask, where the white and light purple pixels are flagged as 'bright'.

EXPECTED NUMBER OF RECOVERABLE TRANSIENTS IN *TESS* (2018-2024)

Magnitude limit	Kilonova	FELT (fainter limit)	FELT (brighter limit)
$m_{TESS} = 19.0$	$0.38\ ^{+0.78}_{-0.30}$	3.12 ± 1.17	392.60 ± 147.22

REFERENCES

[1] R Ridden-Harper et al., K2: Background Survey – the search for undiscovered transients in Kepler/K2 data, Monthly Notices of the Royal Astronomical Society, Vol. 498, Issue 1, October 2020, Pages 33–43, https://doi.org/10.1093/mnras/staa2247

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